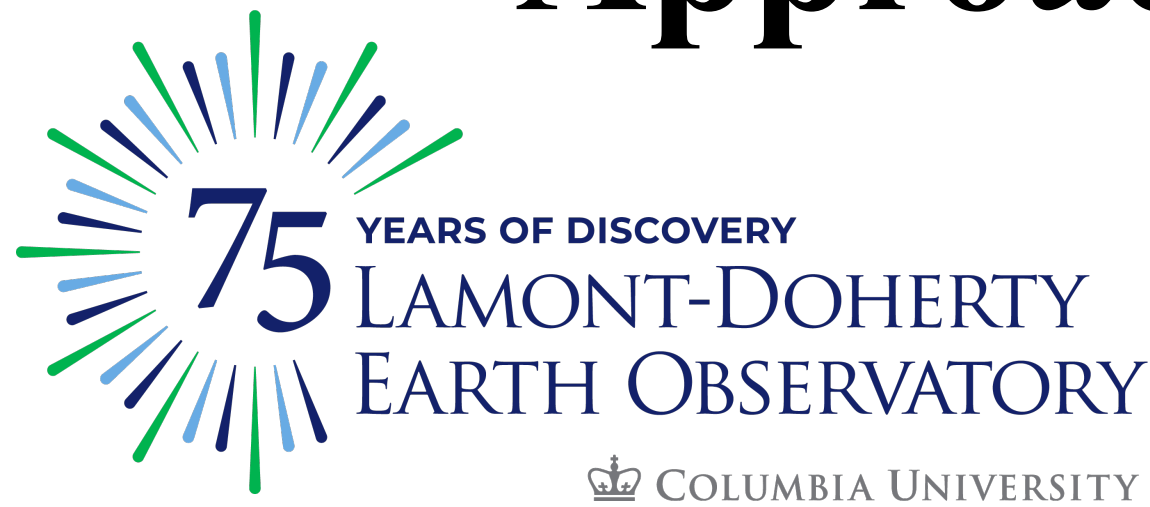


Last Interglacial Sea-Surface Temperature Reconstruction: A Preliminary Approach Using Reduced-Space Optimal Interpolation (130–115 ka)



Anna Chen^{1,2}, Alexey Kaplan³, Jerry McManus^{1,3}

¹ Dept. of Earth and Environmental Sciences, Columbia University
² Dept. of Applied Physics and Applied Mathematics, Columbia University
³ Lamont-Doherty Earth Observatory, Columbia University



COLUMBIA UNIVERSITY
DATA SCIENCE INSTITUTE

Introduction

The Last Interglacial (LIG, MIS5e, ~130–115 ka) was the most recent interval when global temperature was warmer than today. LIG's sea-surface temperature (SST) variability offers a natural analogue for evaluating ocean–climate responses under Anthropocene warming. This study contributes to the collaborative research “*How much and why did Ice Sheets melt during the Last Interglacial?*” (HISEAS) by quantitatively understand SST patterns of the LIG.

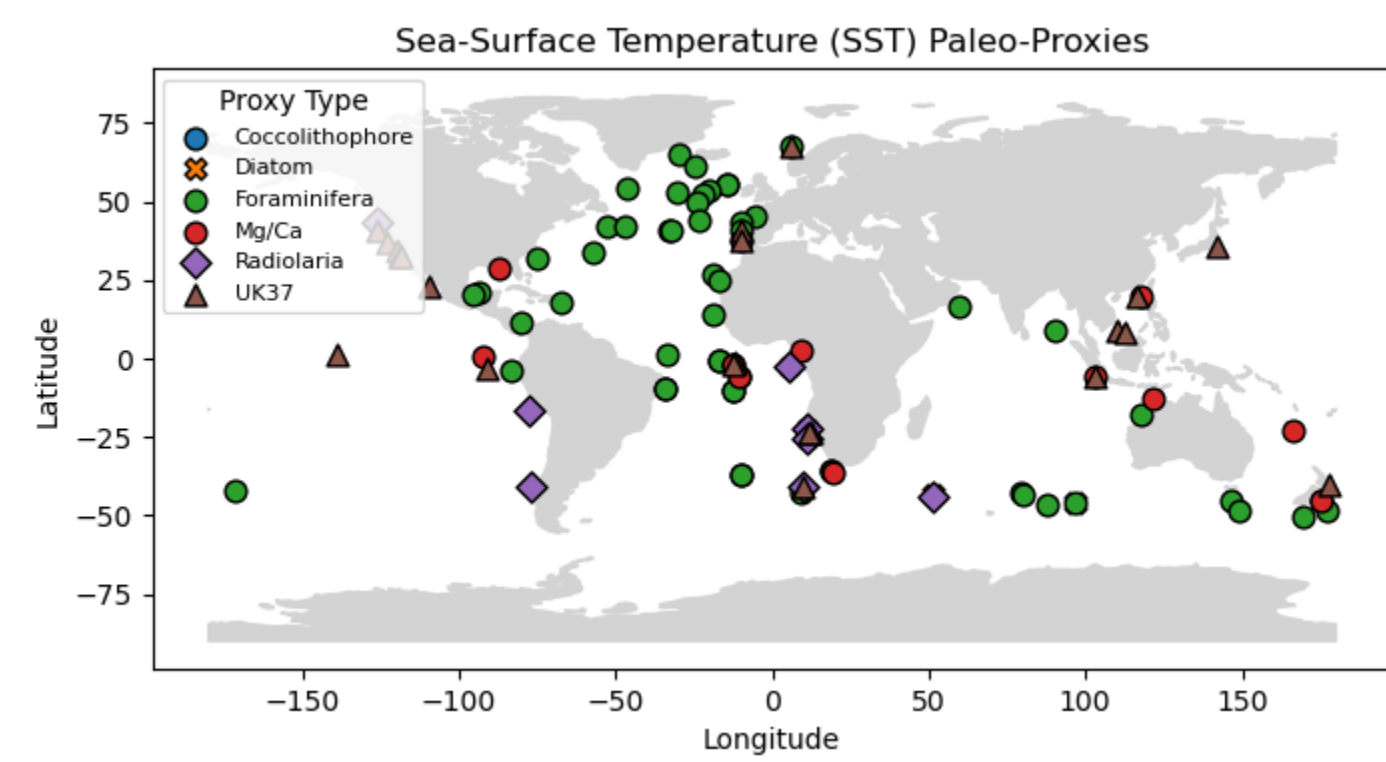
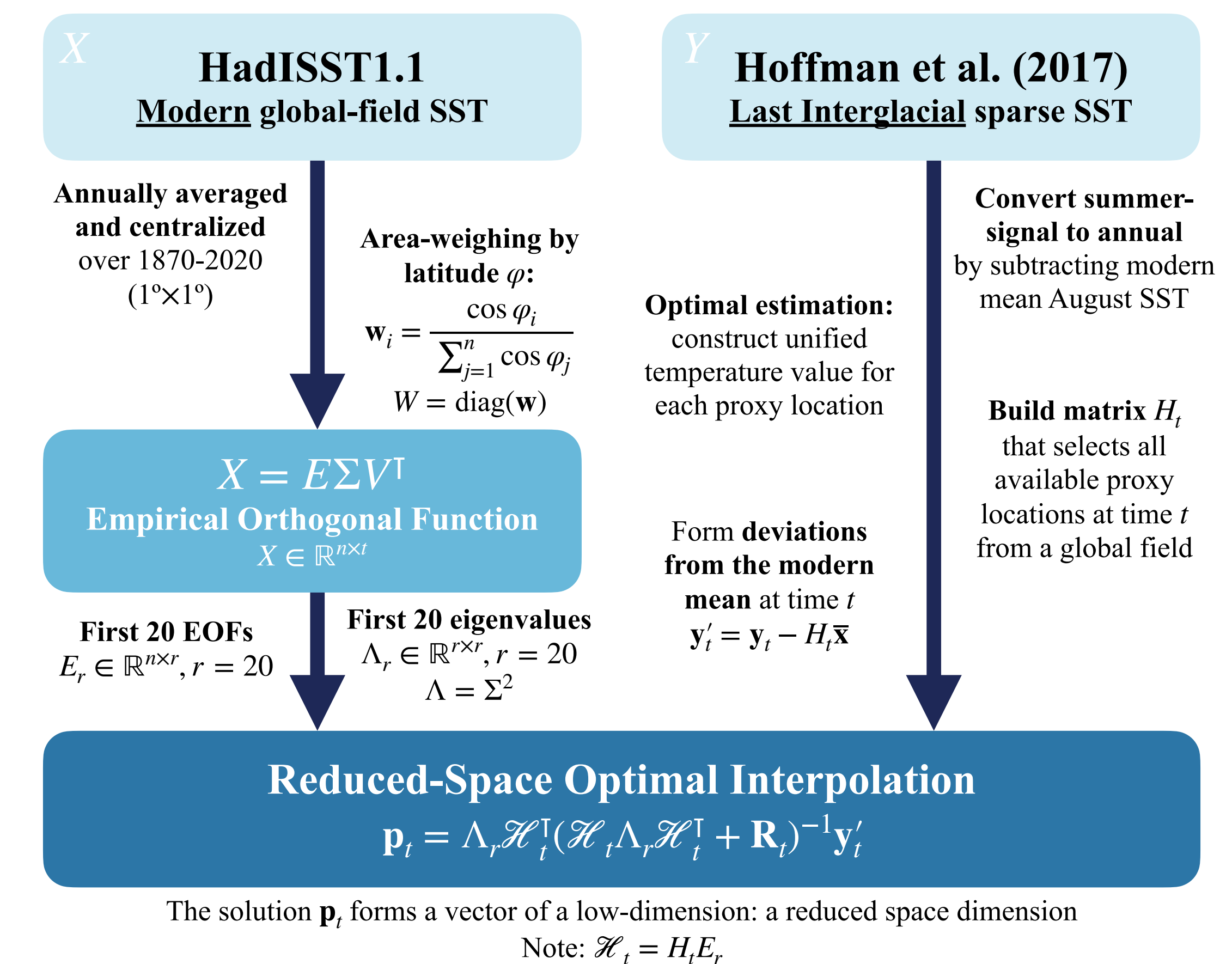


Fig. 1. SST paleo-proxies provided by the Hoffman et al. (2017) dataset.

Given the Hoffman et al. (2017) time-series of 104 paleo-proxy-inferred SST with 0.1 ka intervals, shown in Fig. 1, this study aims to **reconstruct and animate global-field SST** at every 0.1 ka, visualizing the surface ocean forcing patterns and inter-basin surface heat dynamics of the LIG. The key challenge is to assess and limit reconstruction uncertainty while utilizing **spatially sparse and unevenly distributed** paleo SST data.

Methodology



Error Analysis

$$\mathbf{R}_t = \mathbb{E}(\epsilon_t^{obs} (\epsilon_t^{obs})^T) + \mathbb{E}(\epsilon_t^{red} (\epsilon_t^{red})^T)$$

Observational & SST transfer function error covariance matrix
 σ_i is standard deviation at site i supplied by Hoffman et al. (2017)

Rank-reduction error
 from full-rank to the reduced space.
 Tilde represents the unused vectors the truncated eigenvectors and RSOI solutions beyond rank $r = 20$.

$$\begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \\ & & & \sigma_r^2 \end{bmatrix} = \mathbf{H}_t \tilde{\mathbf{E}} \tilde{\mathbf{p}}_t$$

Error reported at each t as the standard deviation with area-weighting undone at site i : $\text{std}_i = \frac{\sqrt{\text{diag}(\mathbf{R}_t)}}{\sqrt{w_i}}$

Global-field SST Reconstruction at each t :

$$\mathcal{T}_t = W^{-\frac{1}{2}} E_r \mathbf{p}_t$$

RSOI constructs the global-field solution ($\mathcal{T}_t$) as a weighed sum of the rank-reduced EOFs (E_r): weighed by its coefficient ($\mathbf{p}_t$) and un-doing area-weighting ($W^{-\frac{1}{2}}$) at each t

Results and Analysis

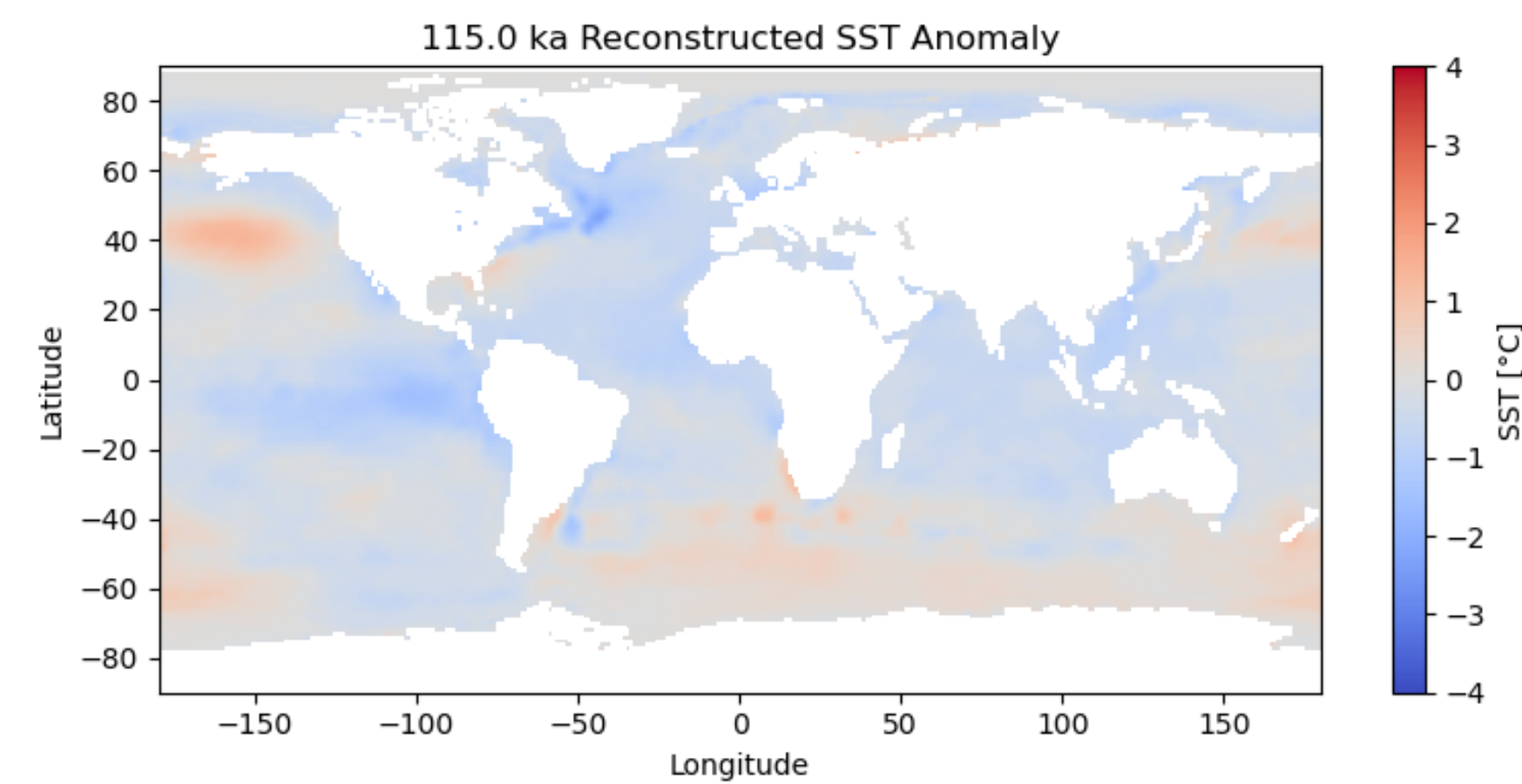


Fig. 2. Global-field SST reconstruction at 115.0 ka. Anomaly from 1870–2020.

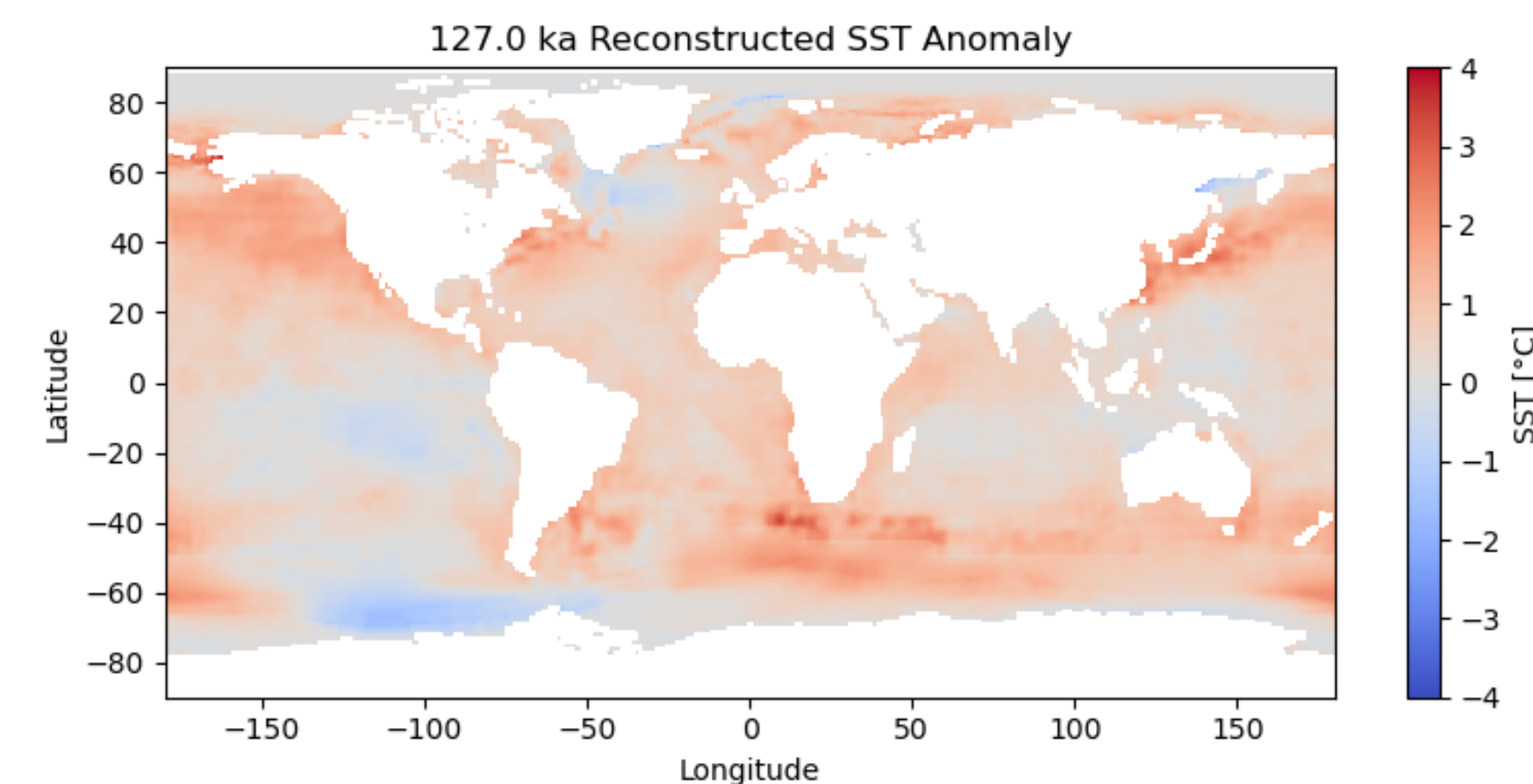


Fig. 3. Global-field SST reconstruction at 127.0 ka. Anomaly from 1870–2020.

- ▶ **N. Atlantic** remains cooler or equal to modern temperature throughout LIG
- ▶ **N. Pacific, S. Pacific, and S. Atlantic** reveal strongest warming up to +4°C
- ▶ Early LIG warming response **first observed ~40°S and the entire Atlantic**, indicating high Atlantic Ocean sensitivity and possible attribution to AMOC
- ▶ Animation highlights **heterogeneous ocean basin responses** and unresolved lead–lag structure across basins

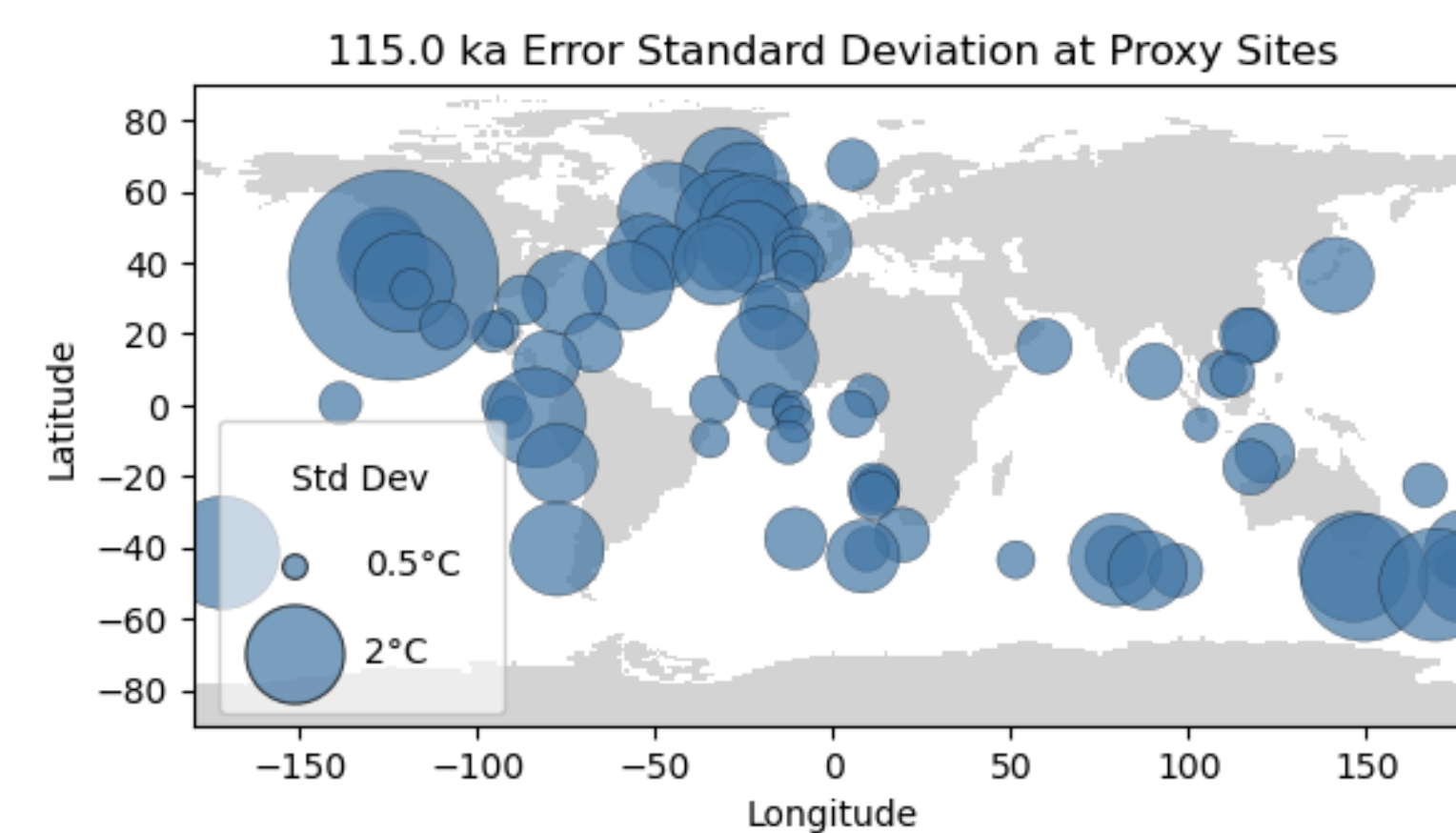


Fig. 4. Reconstruction error (standard deviation) at 115.0 ka at paleo-proxy sites, shown in circle size.

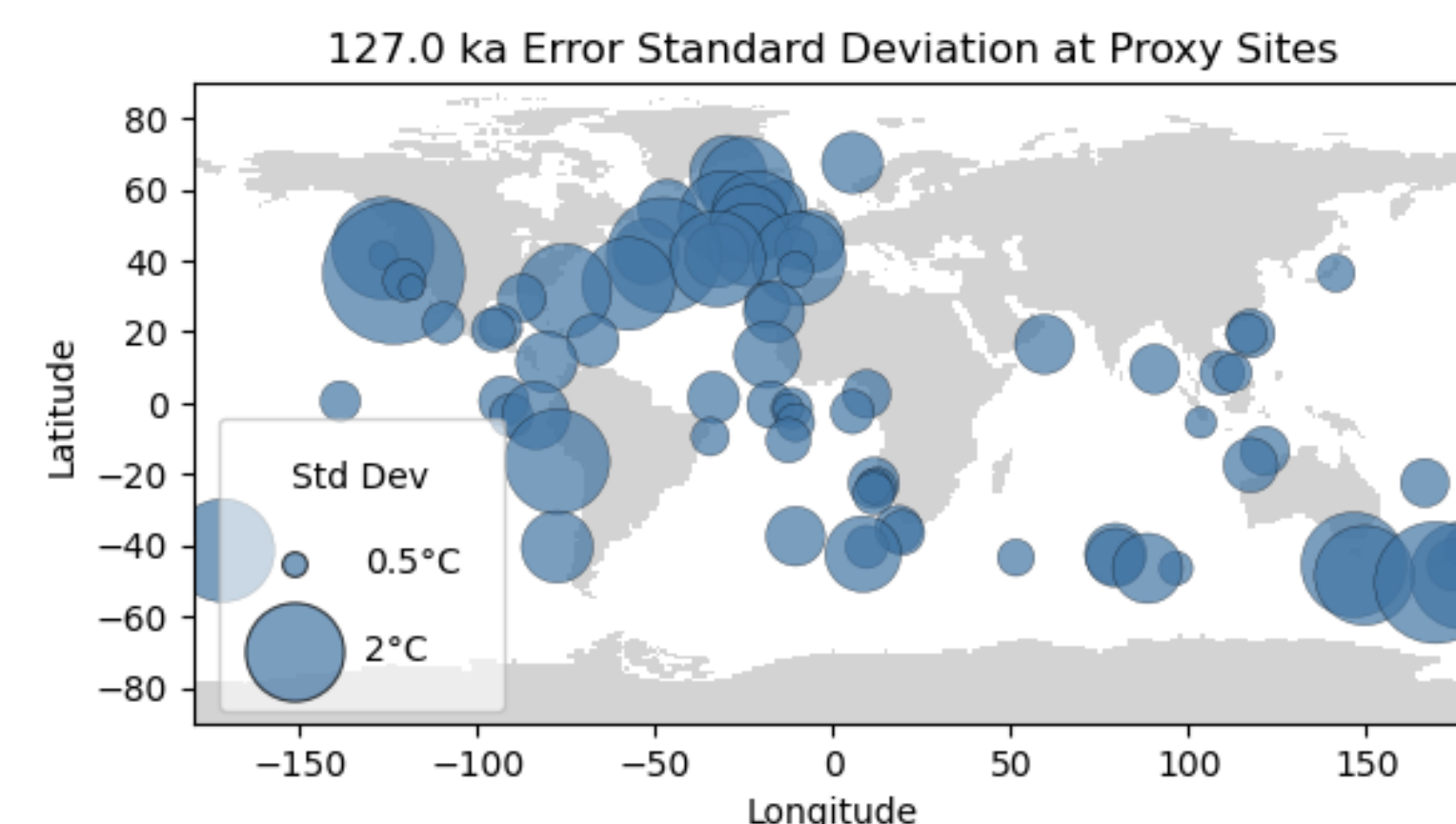


Fig. 5. Reconstruction error (standard deviation) at 127.0 ka at paleo-proxy sites, shown in circle size.

- ▶ First 20 eigenvalues explained **88%** of total variance
- ▶ Largest errors seen in **early LIG (>127 ka)** in **N. Atlantic** and in **late LIG (<117 ka)** in **N.E. and S. Pacific**, up to 4.5 °C
- ▶ Since no paleo-proxy data exist beyond ~50°S, evaluations of the Southern Ocean cannot be conclusively made

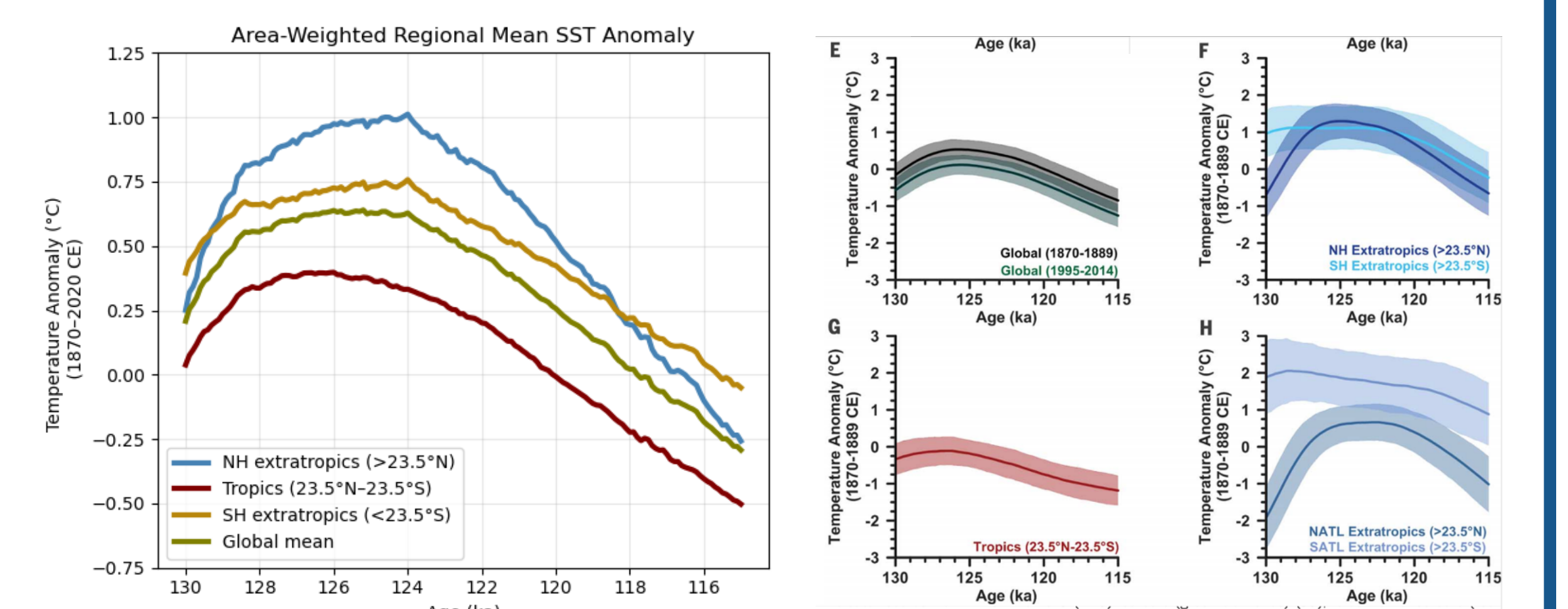


Fig. 6. Global and regional 1°×1° gridded, area-weighted reconstructed SST anomaly stack (this study). Anomaly relative to 1870–2020 mean.

Fig. 7. 5°×5° gridded, area-weighted proxy-based SST anomaly stack from sparse paleo-proxy sites (Hoffman et al., 2017).

- ▶ Variations generally **agree with existing literature**: rapid warming at early-LIG (130–128 ka), maximal SST reached around 126 ka, and near-modern global mean reached by 118 ka
- ▶ Shape of each latitude band mean and the global mean presents good fit
- ▶ **N. Hemisphere extratropics** present the most drastic warming and cooling, and highest temperature anomaly. Conversely, **tropical** anomalies are mild.

Conclusion and Future Work

We implemented Reduced-Space Optimal Interpolation (RSOI) on sparse Last Interglacial (LIG) SST data, using 1870–2020 as the modern baseline for Empirical Orthogonal Function (EOF) analysis. An animation of 1°×1° global-field SST anomaly reconstruction at 0.1 ka intervals is produced. Our preliminary results support previous interpretation on LIG spatial warming and cooling trends, but using a much more statistically robust framework. RSOI was able to compute a high-resolution solution cheaply and also makes it well-constrained. This study shows possibilities of leveraging modern gridded products for spatial paleoclimate reconstruction with sparse data.

Future work should strive to resolve the following issues:

- ▶ Minimize proxy signal uncertainty: structurally align **summer-signal** with annual signals and **calibrate temperature sensitivity** across proxy types.
- ▶ Evaluate whether LIG variations can be described sufficiently well by the dominant patterns found in the **modern annual-scale data**
- ▶ Improve LIG proxy sampling in **central Pacific and Southern Ocean**
- ▶ Assemble proxy-based SST LIG time-series beyond Hoffman et al. (2017)

Acknowledgements

This project is generously sponsored by the Spring 2025 *Data Science Institute Scholars* program at Columbia University, and the Summer 2025 *Heinrich Family Global Fellowship*, a part of the *Columbia College Summer Research Fellowship* overseen by the Undergraduate Research and Fellowships office. We sincerely thank all HISEAS colleagues for their critical feedback and warm encouragements.

Works Cited

- Evans, M. N., Kaplan, A., & Cane, M. A. (2002). Pacific sea surface temperature field reconstruction from coral $\delta^{18}\text{O}$ data using reduced space objective analysis. *Paleoceanography*, 17(1), 7–17–13. <https://doi.org/10.1029/2000PA000590>
- Hoffman, J. S., Clark, P. U., Parnell, A. C., & He, F. (2017). Regional and global sea-surface temperatures during the last interglaciation. *Science*, 355(6322), 276–279. <https://doi.org/10.1126/science.1248464>
- Kaplan, A., Kushnir, Y., Cane, M. A., & Blumenthal, M. B. (1997). Reduced space optimal analysis for historical data sets: 136 years of Atlantic sea surface temperatures. *Journal of Geophysical Research: Oceans*, 102(C13), 27835–27860. <https://doi.org/10.1029/97JC01734>
- Rayner, N. A., Parker, D. E., Horton, E. B., Folland, C. K., Alexander, L. V., Rowell, D. P., et al. (2003). Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. *Journal of Geophysical Research: Atmospheres*, 108(D14). <https://doi.org/10.1029/2002JD002670>

1. Modern Data Empirical Orthogonal Function (EOF)

We implement Reduced-Space Optimal Interpolation (RSOI) on the Hoffman et al. (2017) compiled data set of paleoclimatic proxy records. Annually-averaged 1°-by-1° SST fields from the U.K. Hadley Centre’s HadISST1 data set for the 1870–2020 period are area-weighted by $\mathbf{E}^T \mathbf{W} \mathbf{E} = \mathbf{I}$ such that the area-weighted EOF, $\mathbf{E}$, is orthogonal. Then, Empirical Orthogonal Function (EOF) analysis is performed as:

$$\mathbf{X} = \mathbf{E} \mathbf{\Sigma}^T \mathbf{V}.$$

2. LIG Proxy Data Interpolation and Area-weighting

Optimal estimation of LIG proxy record time-series ensured a single temperature value is produced on 0.1 ka intervals. Rank-20 reduced EOFs form the basis of a low-dimensional linear space (“reduced space”) within which the solution (an interpolated SST field) is sought for each specified time value, based on the proxy records available for that given time.

3. Reduced-Space Optimal Interpolation (RSOI)

RSOI constructs the SST field solution as a weighed sum of the EOF patterns, each weighed by its own coefficient. These coefficients form a vector of a low-dimension — a reduced space dimension — found using a low-dimensional Optimal Interpolation (OI) procedure:

$$\mathbf{p}_t = \mathbf{\Lambda} \mathcal{H}_t^T (\mathcal{H}_t \mathbf{\Lambda} \mathcal{H}_t^T + \mathbf{R}_t)^{-1} \mathbf{y}_t$$

where $\mathcal{H}_t = \mathbf{H}_t \mathbf{E}$ is the factor of a time-variant selector matrix $\mathbf{H}_t$ that selects the location of the proxies at the given time t and the matrix of eigenvector $\mathbf{E}$, $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues of the spatial covariance matrix $\mathbf{C}$ from EOF analysis ($\mathbf{C} = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^T$), $\mathbf{y}_t$ is the paleo-proxy SST data vector at t .

4. Error Analysis

The error matrix, $\mathbf{R}_t$, is constructed from the sum of two components:

$$\mathbf{R}_t = \mathbb{E}(\mathcal{G}_t^{red}(\mathcal{G}_t^{red})^T) + \mathbb{E}(\mathcal{G}_t^{obs}(\mathcal{G}_t^{obs})^T)$$

where $\mathbb{E}(\mathcal{G}_t^{obs}(\mathcal{G}_t^{obs})^T) = \text{diag}(\begin{bmatrix} \sigma_1^2 & \sigma_2^2 & \dots & \sigma_{n_t}^2 \end{bmatrix})$ is the standard deviation

reported by the Hoffman et al. (2017) dataset from observational and transfer function errors. $\mathbb{E}(\mathcal{G}_t^{red}(\mathcal{G}_t^{red})^T) = \mathbf{H}_t \tilde{\mathbf{E}} \tilde{\mathbf{p}}_t$ is the error of truncation of the full-dimensional SST field to its reduced space, where tilde represents the unused eigenvectors and RSOI solutions beyond Rank 20.

5. Reconstruct the Global Field

The global field SST is reconstructed via the RSOI solution and un-doing area-weighting at each timestep t :

$$\mathcal{T}_t = \mathbf{W}^{-\frac{1}{2}} \mathbf{E} \mathbf{p}_t$$

Animation of reconstructed global-field SST at every 0.1 ka time-steps over 115.0-130.0 ka is produced.

Evans, M. N., A. Kaplan, and M.A. Cane, 2002: Pacific sea surface temperature field reconstruction from coral d18O data using reduced space objective analysis, *Paleoceanography*, **17**(1), 10.1029/2000PA000590.

Hoffman, J. S., Clark, P. U., Parnell, A. C., & He, F. (2017). Regional and global sea-surface temperatures during the last interglaciation. *Science*, 355(6322), 276–279. <https://doi.org/10.1126/science.aai8464>.

Kaplan, A., Y. Kushnir, M. Cane, and M. Blumenthal, 1997: Reduced space optimal analysis for historical datasets: 136 years of Atlantic sea surface temperatures, *J. Geophys. Res.*, **102**, 27835-27860.

Rayner, N. A., Parker, D. E., Horton, E. B., Folland, C. K., Alexander, L. V., Rowell, D. P., et al. (2003). Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. *Journal of Geophysical Research: Atmospheres*, 108(D14). <https://doi.org/10.1029/2002JD002670>